Source Transformation

Concept and Definition

Source transformation is a technique used in circuit analysis to convert a voltage source in series with a resistor into an equivalent current source in parallel with the same resistor, and vice versa.

Both circuits are electrically equivalent, meaning they produce the same voltage and current at their terminals.

```
Voltage Source → Current Source I = V / R
```

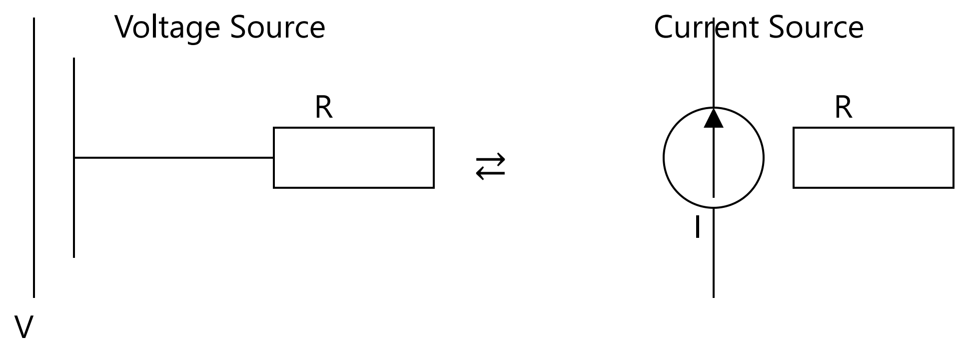
```
Current Source → Voltage Source V = I × R
```

Why Source Transformation Works

The transformation is based on the fact that both forms deliver the same terminal behavior. That means the voltage across and current through any load connected to the terminals remains unchanged.

This concept is closely related to Thevenin's and Norton's theorems.

Basic Transformation Diagrams



Step-by-Step Transformation

Voltage Source → Current Source

1. Identify voltage source (V) in series with resistance (R) 2. Convert to current source using:

> I = V / R

3. Place the same resistance in parallel with the current source

Current Source → Voltage Source

1. Identify current source (I) in parallel with resistance (R) 2. Convert using:

> V = I × R

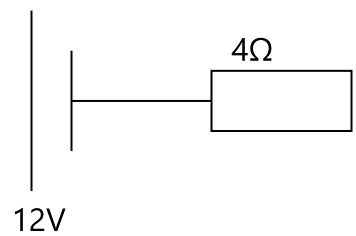
3. Place the same resistance in series with the voltage source

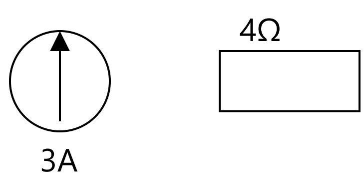
Example 1: Voltage to Current Conversion

A 12V voltage source is connected in series with a 4Ω resistor.

```
I = 12 / 4 = 3 A
```

Equivalent circuit: 3A current source in parallel with 4Ω resistor.





Example 2: Current to Voltage Conversion

A 2A current source is in parallel with a 5Ω resistor.

```
V = 2 × 5 = 10V
```

Equivalent circuit: 10V voltage source in series with 5Ω resistor.

When to Use Source Transformation

Source transformation is especially useful when simplifying circuits before applying nodal or mesh analysis.

It should be used:

- When circuits contain mixed voltage and current sources
- When simplifying before applying Thevenin or Norton theorem
- When reducing complex networks into simpler forms

It should not be used blindly. The transformation must preserve the original circuit behavior.

Relation to Other Methods

Source transformation works alongside:

- **Nodal Analysis** – simplifies node equations
- **Mesh Analysis** – reduces loop complexity
- **Thevenin/Norton Theorem** – helps find equivalents faster

Using these methods together makes solving circuits significantly more efficient.

Practical Applications

Source transformation is widely used in electrical engineering to simplify circuits before analysis. It is particularly useful in power systems, electronics design, and troubleshooting complex networks.

Engineers often use it to convert circuits into forms that are easier to analyze using standard techniques.

Source Transformation

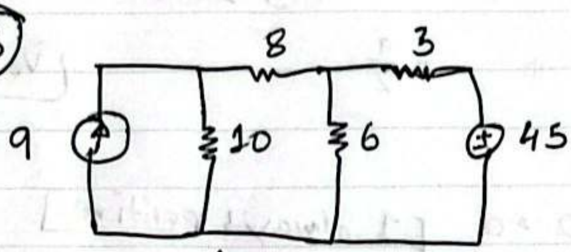
$V = IR$

From current to voltage
parallel → series

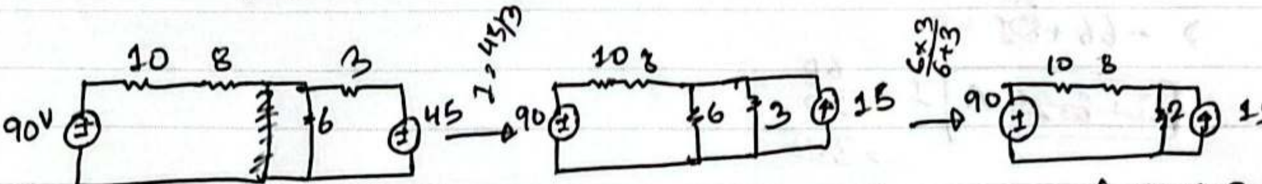
$I = \frac{V}{R}$ [I always positive in loop]

From voltage to current
series → parallel

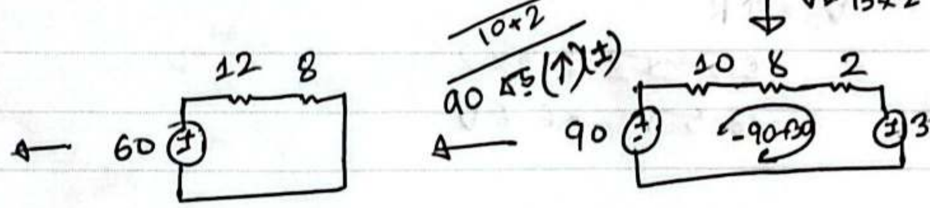
(4.23) ∴ Current & power in 8Ω



↓ $V = 9 \times 10$



↓ $V = 15 \times 2$



$\Rightarrow I = \frac{V}{R}$

$= \frac{60}{12+8}$

$\boxed{I = 3A}$ [Current is same in series circuit]

$P = I^2 R$

$= 3^2 \times 8$

$\boxed{P = 72W}$

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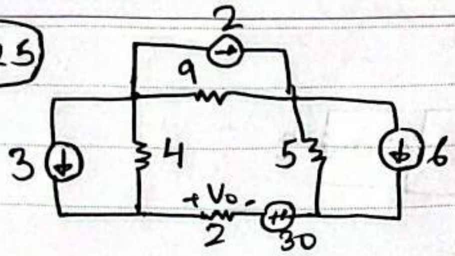
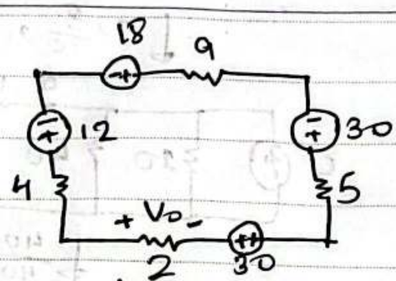
(4.25)



$\longrightarrow$



∴ Obtain V_o

KVL: [current +]

$-18+9I-30+5I-30+2I+4I+12=0$

$\Rightarrow -66+20I=0$

$\Rightarrow I=\frac{66}{20}$

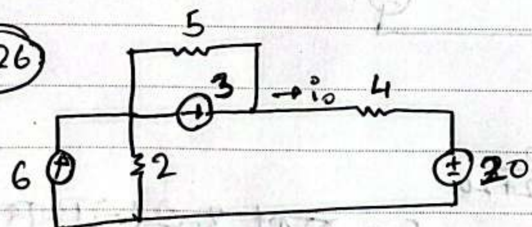
$I=-3.3$ [- কারণ, V_o এ (-) থেকে যাচ্ছে]

$\therefore V_o = IR$

$= 3.3\times 2$

$\boxed{V_o = -6.6\ V}$

(4.26)

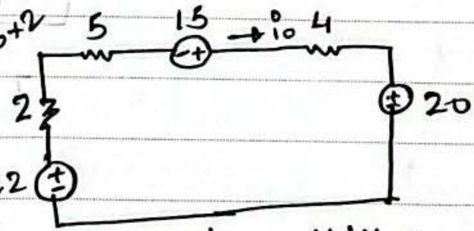


∴ Find i_o

$V=5\times3$

$V=6+2$

$\longrightarrow$



KVL:

$4i_o+20-12+2i_o+5i_o-15=0$

$\Rightarrow -7+11i_o=0$

$\Rightarrow i_o=\frac{7}{11}$

$\boxed{i_o = 0.63\ A}$

FIGURE NO. ....

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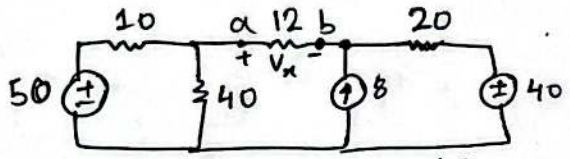
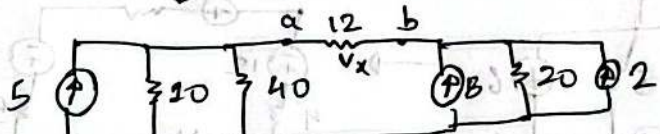
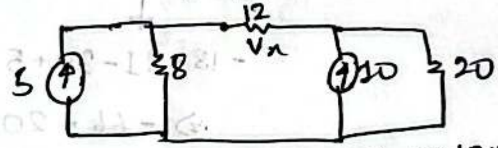


$\therefore$ find V_x

$\downarrow$ $I = \frac{50}{10}$, $I = \frac{40}{20}$

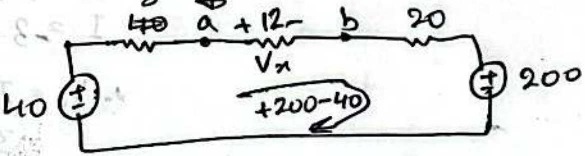


$\downarrow$ $\frac{40\times10}{40+10}$, $8+2$

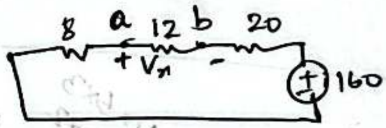


$V = 8\times5$, $V = 10\times20$

8 $\downarrow$



$\downarrow$ 200 বড়



$I = \frac{V}{R}$

$= \frac{160}{8+12+20}$

$I = -4$ [- কারণ V_x এ (-) থেকে যাচ্ছে]

$\Rightarrow V_x = IR$

$= 4\times12$

$\boxed{V_x = 48\ V}$